

## Variables

```
In [ ]: # to store the value  
  
# varyyyyyyyyyyyyyyy  
  
# It can also change
```

```
In [1]: a=10  
a  
  
# 10 is a vlaue stored in a variable a
```

Out[1]: 10

```
In [2]: a=20  
a  
  
# 20 is a value stored in a variable a
```

Out[2]: 20

```
In [3]: number=100  
number
```

Out[3]: 100

```
In [4]: NUMBER=200  
NUMBER
```

Out[4]: 200

```
In [6]: number,NUMBER
```

Out[6]: (100, 200)

```
In [7]: NUMber=300  
NUMber
```

Out[7]: 300

```
In [8]: Num123=400  
Num123
```

Out[8]: 400

```
In [9]: 123num=500
        123num
```

```
Cell In[9], line 1
      123num=500
      ^
SyntaxError: invalid decimal literal
```

```
In [10]: 500=600
```

```
Cell In[10], line 1
      500=600
      ^
SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of
'='?
```

```
In [11]: number$=700
        number$
```

```
Cell In[11], line 1
      number$=700
      ^
SyntaxError: invalid syntax
```

```
In [12]: number=$700
```

```
Cell In[12], line 1
      number=$700
      ^
SyntaxError: invalid syntax
```

```
In [13]: number@hello=800
        number@hello
```

```
Cell In[13], line 1
      number@hello=800
      ^
SyntaxError: cannot assign to expression here. Maybe you meant '==' instead of
'='?
```

```
In [14]: number_=900
        number_
```

```
Out[14]: 900
```

```
In [15]: _=1000
_
```

Out[15]: 1000

```
In [16]: sum=900
sum
```

Out[16]: 900

```
In [ ]: in python green colour means keywords
math keywords works,remaining all other keywords fail
```

```
In [17]: max=600
max
```

Out[17]: 600

```
In [18]: if=900
```

```
Cell In[18], line 1
    if=900
      ^
SyntaxError: invalid syntax
```

```
In [19]: for=800
```

```
Cell In[19], line 1
    for=800
      ^
SyntaxError: invalid syntax
```

```
In [20]: sum=900
```

```
In [ ]: SUM=900
```

```
In [23]: AGE=30
AGE
age=50
age

AGE,age
```

Out[23]: (30, 50)

- Balck colour: variables
- Variables can be upper case: NUMBER
- Variables can be lower case : number
- Variables can be combination of Upperand Lower : NUMber
- Letters and Numbers as suffix combination works: number123

- underscore can use as variables
- Numbers as prefix is invalid: 123number
- spaces , spl charcaters ,symbols are invalid : number\$, number@
- Green color: keywords
- Other than math keywords are in valid
- sum max min math keywords

In [24]: number one=200

Cell In[24], line 1

number one=200

^

**SyntaxError:** invalid syntax

In [25]: number\_one=200

In [ ]: